

ARSUM NADEEM CHAUDHARY

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AI software engineer with experience in distributed systems, scalable infrastructure, and AI-driven automation. Built high-impact production systems at a YC-backed startup, including Mulligan (a product later acquired).

EDUCATION

Pomona College | Claremont, CA (*Full Financial Aid Scholar*)

May 2025

Bachelors of Arts in Computer Science and Mathematics (Double Major)

GPA: 3.8/4.0

Coursework: Data Structures, Algorithms, Big Data, Complex Systems, AI, Neural Nets, Operations Research, Adv Linear Algebra

SKILLS

Languages: Python, JavaScript, Java, SQL

Systems & Infrastructure: Distributed systems, REST APIs, AWS (EC2, S3), GCP, Docker, Kubernetes, database design & data modeling, React, LLM applications, Prompt Engineering, AI Product

EXPERIENCE

Software Engineer (AI) - PyqAI (YC W23) | [Mulligan](#) (Acquired)

June 2025 - Mar 2026

- Automated **~90%** of insurance back-office workflows by building data pipelines and dockerized RPA services with AI fall back, enabling **200+** brokers to quote policies **6x faster** (<10 min vs 1 hr) and saving clients **~\$700K** annually
- Reduced end-to-end latency **10x (150s → 15s)** by identifying demo bottlenecks using user behavior data and parallelizing distributed cloud functions across AWS and GCP
- Built **self-serve backend** and cloud infrastructure for Mulligan, including third-party API integrations and a usage-based billing system (Stripe + Metronome) to support **production SaaS workflows**
- Engineered multi agentic orchestration to integrate unreliable insurance carrier APIs, handling retries, schema normalization, partial failures, and ensuring consistent end-state across requests

Software Engineering Intern - [Learning Economy Foundation](#)

Sep 2023 - Dec 2023

- Streamlined credential generation for **1k+ daily users** by building automated backend preview pipelines using OpenAI function calling, increasing credential issuance by **200%**, improving learner retention by **15%**, and reducing engineering overhead by **10+ hours/week**

Data Engineering Intern - [Inabia Software & Consulting](#)

June 2023 - Aug 2023

- Enabled pharmaceutical clients (e.g., Allergan Aesthetics) to ensure data integrity for compliance audits and boost processing efficiency by **20%** by building ETL pipelines to ingest/normalize **1TB** of sensitive health data
- Built NLP models to flag non-compliant advertising language, reducing compliance review time by **50%** and improving audit readiness and regulator trust

LEADERSHIP

Lead College Application Coach | Pomona College Academy for Youth Success ([PAYS](#))

May 2022 - Aug 2024

- Led team of **15** to coach **92** underrepresented high-school students over **3 summers** in Pomona's intensive three-year equity mentorship program for STEM & college applications, contributing to placements at **top US universities** and **\$760k** secured in financial aid and scholarships
- Taught liberal arts-style critical thinking, communication, and problem-solving skills by creating a first-time **Math and Music module**, preparing students to thrive in selective university environments beyond technical coursework

RELEVANT PROJECTS

Brain Tumor Classification Model - [GitHub](#)

Jan 2025 - Feb 2025

- Enabled real-time tumor classification (**<200ms per image, ~96% accuracy**) by designing a high-throughput pipeline with an Xception + custom CNN ensemble to preprocess 1k+ MRI scans
- Built **saliency-maps** to visualize predictions, improving clinician trust in AI medical decision-making
- Presented the applied AI solution to Google and healthcare leaders at an accelerator program

Twitter Clone Web Application - [GitHub](#)

May 2024 - June 2024

- Built a full-stack social platform supporting **1M+** posts (simulated workload) by designing optimized database schemas, indexed queries, and scalable deployment using Docker, Nginx, and Gunicorn on AWS behind a load balancer
- Implemented **REST APIs** for posts, likes, follows, and authentication to support high-throughput reads and writes

Coronavirus Twitter Analysis Project - [GitHub](#)

Jan 2024 - Feb 2024

- Analyzed **1.1B tweets** in 10+ languages across 20+ countries to create **4 critical visuals**, highlighting cultural and linguistic differences in user reactions and demonstrating the power of data storytelling for empathy at global scale

ADDITIONAL INFORMATION

Involvement: Team Lead: UCLA DataFest, 4x AI Hackathons; College Speech & Debate; Music (Balinese Gamelan, solo flutist)

Honors: Silicon Valley Program Scholar; Pomona College Scholar (multiple semesters, 4.0 GPA); Full Financial Aid International Scholar; Sigma Xi Research Honor Society (Math Thesis); Founded First Library at Human Rights Activist Mukhtar Mai's School